

MTOjects / Toy Objects Optimisation

Four-fold cross-validation on 40 low-foreground galaxies

1. Aim

The MTOjects optimisation calibrates foreground-object masking against known injected Toy Objects while penalising unnecessary removal of galaxy pixels. It repeats the SEP experimental design on the same 40 validated galaxies, allowing the two algorithms to be compared under a common sampling framework.

2. Data and Toy Objects

The input list contains exactly 40 unique galaxies drawn from the 182-galaxy manifest. Each has complete bar/disc geometry and an accessible 3.6-micron FITS image. Six stars, compact clusters or elliptical galaxies are injected per image within the investigated region. Peaks span 5-25 robust-sigma and truth masks include model pixels above 8% of peak followed by one-pixel dilation.

Toy property	Rule
Type mixture	50% star, 20% compact cluster, 30% galaxy
Star/cluster FWHM	2-10 pixels
Galaxy FWHM	5-22 pixels
Galaxy axis ratio	0.35-0.95
Placement	Non-overlapping and wholly inside the investigated region

3. Four-fold design

A fixed shuffle divides the same 40 names used by SEP into four disjoint folds of 10. In each rotation, MTOjects is optimised on 30 galaxies and evaluated on the excluded 10. Every galaxy therefore appears in a held-out set once. Each fold uses 40 Optuna trials: eight startup samples followed by 32 adaptive TPE trials.

Phase	Images	Influences optimisation?
Training	30 per fold	Yes
Held-out validation	10 per fold	No
Common candidate evaluation	All 40, independent injections	Selects among four trained candidates
Final batch	All 182 manifest galaxies	No

4. MTOjects processing model

MTOjects is run on the original science image for Toy Objects calibration. A baseline mask is first computed from the unmodified image. The same parameters are then applied to the injected image, and scoring uses only incremental pixels that were absent from the baseline mask. This separates Toy Object response from objects already detected in the galaxy.

incremental mask = injected-image mask AND NOT baseline mask

5. Objective

Recovery reward: $R_{MT} = 0.45 \text{ mean F-score} + 0.35 \text{ mean toy recall} + 0.20 \text{ toy detection rate}$.

Data-loss term: $L_{MT} = 0.50 \text{ mean masked fraction} + 0.10 \min(\text{false-positive fraction}, 1)$. Feasible score: $S_{MT} = R_{MT} - L_{MT}$; Optuna minimises objective = $-S_{MT}$.

A toy is counted as detected when at least 50% of its truth pixels are recovered. A trial is recovery-infeasible when the incremental mask is empty, toy detection rate is below 25%, or mean toy recall is below 20%; it receives a large positive objective, with an empty mask receiving at least 100. Feasible trials retain the 15% worst-image masking cap. Final release requires common-40 toy detection of at least 50%, common-40 mean toy recall of at least 30%, and non-zero recovery in every held-out fold.

6. Optimised parameters and constraints

The search domains are the established constraints already implemented by the project MTOjects optimiser. Parameters not listed remain fixed at the processing-model defaults.

Parameter	Constraint	Interpretation
move_factor	0.0-1.0	Tree-node movement control
min_distance	0.0-1.0	Minimum hierarchy distance
gaussian_fwhm	0.0-5.0 pixels	Gaussian smoothing applied before segmentation
bg_variance	0.0001-10000; step 0.0001	Background variance supplied to MTOObjects
minarea	1-80 pixels	Minimum retained segment area
dilation_radius	0-8 pixels	Expansion of the accepted foreground mask
max_area	20-3000 pixels	Upper area gate protecting galaxy structure
max_elongation	1.5-20	Shape gate rejecting extreme segments

7. Candidate evaluation and selection

Each fold contributes its best training parameter set. That candidate is scored on the fold's held-out 10 galaxies and on a common independent Toy Object realization across all 40 galaxies. The common set removes differences caused by comparing candidates on different held-out galaxies. The best feasible common-40 candidate is selected for the final batch only after the cross-fold recovery release gates are satisfied.

8. Outputs

Artifact	Review purpose
Fold trial summaries	Inspect every tested parameter combination and objective component
Held-out details	Review per-galaxy generalisation and failures
Candidate comparison	Compare training, held-out and common all-40 metrics

Artifact	Review purpose
Best JSON	Preserve winning parameters and full provenance
182-galaxy summary	Confirm PNG status, runtime, mask fraction and errors

9. Restart and monitoring

The workflow stores every Optuna study in SQLite and writes fold summaries incrementally. Supervisors detect unexpected launcher exits, reuse folds with all 40 trials, and restart incomplete folds or batches. Batch resume mode skips galaxies already present in the summary. Progress, remaining trials, rough ETA and expected completion are retained in terminal logs.

10. Interpretation

The preferred solution should combine stable parameters across folds with strong held-out recovery, controlled false-positive masking and no repeated dependence on extreme parameter bounds. The common 40-galaxy evaluation uses new injected objects but not a wholly external galaxy sample, so it measures injection-level generalisation within the selected low-foreground population.

Systematic documentation review - 16 August 2026

Review classification: Updated - recovery objective, release gates and final winner

Updates made: The obsolete soft score was replaced by the implemented recovery reward/loss, infeasibility penalty and final release gates. Fold-3 selected parameters and common-40 metrics were recorded.

Current interpretation: MTOObjects is no longer a zero-mask solution. It is a feasible secondary method, but mean precision of 0.31% and false-positive fraction of 9.05% prevent recommendation as the primary production mask.

Authoritative combined outcome: SEP and MTOObjects Toy Objects Optimisation Summary and Conclusions.docx. Full-precision results: Foreground Masking Optimisation Results.xlsx.

Selected fold: 3. Common-40 score: 0.2325; mean toy recall: 50.7%; toy detection rate: 53.3%; mean masked fraction: 9.07%; maximum masked fraction: 14.63%; false-positive fraction: 9.05%; mean pixel precision: 0.31%.

Selected parameters: detect_on=original; move_factor=0.805164; min_distance=0.215158; gaussian_fwhm=0.429846; bg_variance=0.000821033; minarea=5; dilation_radius=3; max_area=2987; max_elongation=12.3811; exclude_center_pixels=8; alpha=1e-6.